

SHETH L.U.J AND SIR M.V COLLEGE

SUBJECT - CYBER SECURITY

PRACTICAL NO 7 B

AIM - Configure and Implement Secure Web Communication Using SSL/TLS

Step 1: Update the System

Step 2: Install Apache Web Server

```
mvl@IT-12: /etc/sn/MySQL$ sudo apt update
[sudo] password for mvl:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [967 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [205 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1201 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1227 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [240 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.3 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1385 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [320 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [286 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1687 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [337 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1483 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [339 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [12.8 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:23 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5768 B]
Get:24 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.6 kB]
Fetched 10.9 MB in 6s (1855 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
85 packages can be upgraded. Run 'apt list --upgradable' to see them.
mvl@IT-12: /etc/sn/MySQL$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following NEW packages will be installed:
  ethtool
The following packages will be upgraded:
  apparmor apport-core-dump-handler apt apt-utils base-files binutils
  binutils-common binutils-x86-64-linux-gnu cloud-init console-setup
  console-setup-linux coreutils dconf-gsettings-backend dconf-service dhcpd-base
```

Step 3: Install OpenSSL

```
mvl@IT-12: /etc/sn/MySQL$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3
  libaprutil1-ldap libaprutil1t64 liblua5.4-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser ufw
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3
  libaprutil1-ldap libaprutil1t64 liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 0 not upgraded.
Need to get 2095 kB of archives.
After this operation, 8129 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [11.2 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9116 B]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-3build2 [166 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.15 [1338 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.15 [163 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.15 [99.7 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu8.15 [90.2 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Fetched 2095 kB in 8s (2639 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libapr1t64:amd64.
(Reading database ... 41258 files and directories currently installed.)
Preparing to unpack .../0-libapr1t64_1.7.2-3.1ubuntu0.1_amd64.deb ...
Unpacking libapr1t64:amd64 (1.7.2-3.1ubuntu0.1) ...
Selecting previously unselected package libaprutil1t64:amd64.
Preparing to unpack .../1-libaprutil1t64_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1t64:amd64 (1.6.3-1.1ubuntu7) ...
Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
Preparing to unpack .../2-libaprutil1-dbd-sqlite3_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1-dbd-sqlite3:amd64 (1.6.3-1.1ubuntu7) ...
```

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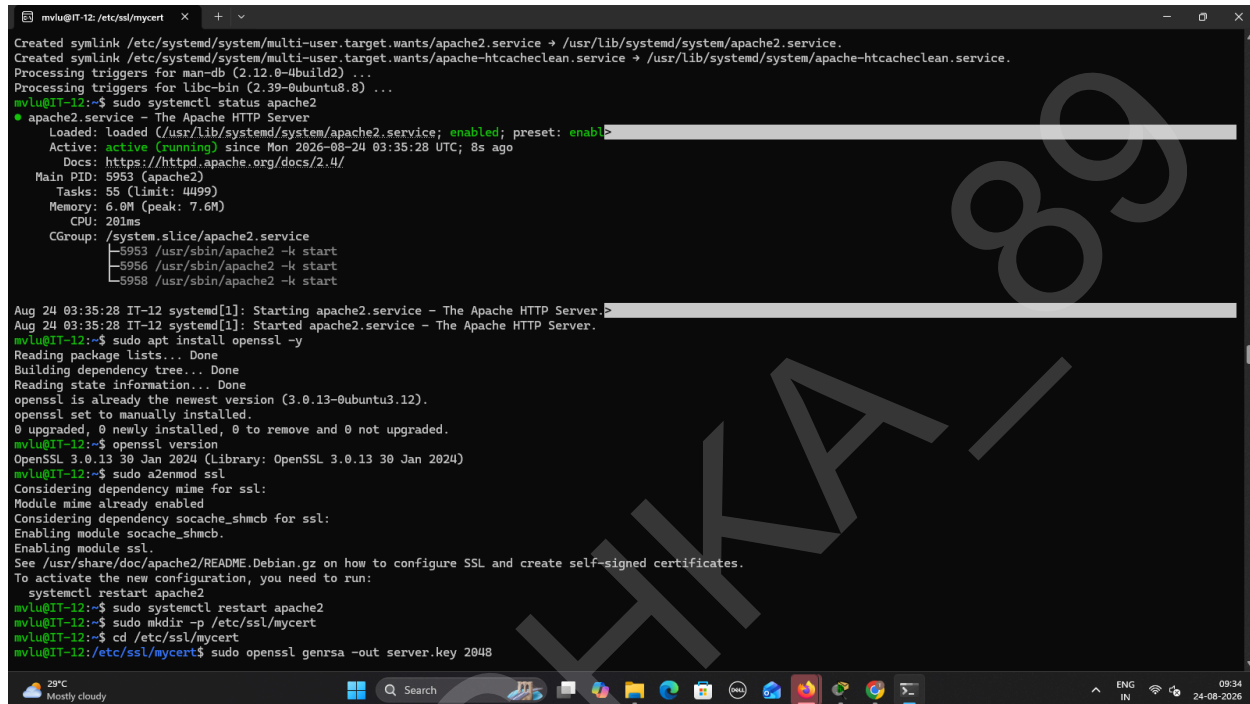
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Step 4: Enable SSL Module in Apache

Step 5: Create a Directory for Certificates



```
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
Processing triggers for man-db (2.12.0-ubuild2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.8) ...
mvl@IT-12:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-08-24 03:35:28 UTC; 8s ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 5953 (apache2)
      Tasks: 55 (limit: 4499)
     Memory: 6.0M (peak: 7.6M)
        CPU: 201ms
    CGroup: /system.slice/apache2.service
            └─5953 /usr/sbin/apache2 -k start
              └─5956 /usr/sbin/apache2 -k start
                └─5958 /usr/sbin/apache2 -k start

Aug 24 03:35:28 IT-12 systemd[1]: Starting apache2.service - The Apache HTTP Server.
Aug 24 03:35:28 IT-12 systemd[1]: Started apache2.service - The Apache HTTP Server.
mvl@IT-12:~$ sudo apt install openssl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssl is already the newest version (3.0.13-0ubuntu3.12).
openssl set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
mvl@IT-12:~$ openssl version
OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
mvl@IT-12:~$ sudo a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
  systemctl restart apache2
mvl@IT-12:~$ sudo systemctl restart apache2
mvl@IT-12:~$ sudo mkdir -p /etc/ssl/mycert
mvl@IT-12:~$ cd /etc/ssl/mycert
mvl@IT-12:~$ sudo openssl genrsa -out server.key 2048
```

Step 6: Generate a Private Key

Step 7: Generate a Self-Signed SSL Certificate

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```
mvl@IT-12: /etc/ssl/mycert$ mvlu@IT-12: /etc/ssl/mycert$
Country Name (2 letter code) [AU]:IN
State or Province Name (full name) [Some-State]:Maharashtra
Locality Name (eg, city) []:Mumbai
Organization Name (eg, company) [Internet Widgits Pty Ltd]:My College
Organizational Unit Name (eg, section) []:CS Department
Common Name (e.g. server FQDN or YOUR name) []:localhost
Email Address []:rathodunnatics242541@gmail.com
mvlu@IT-12: /etc/ssl/mycert$ sudo a2ensite default-ssl
Enabling site default-ssl.
To activate the new configuration, you need to run:
    systemctl reload apache2
mvlu@IT-12: /etc/ssl/mycert$ sudo nano /etc/apache2/sites-available/default-ssl.conf
mvlu@IT-12: /etc/ssl/mycert$ sudo apache2ctl configtest
Syntax OK
mvlu@IT-12: /etc/ssl/mycert$ sudo systemctl restart apache2
mvlu@IT-12: /etc/ssl/mycert$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enable>
   Active: active (running) since Mon 2026-08-24 03:46:55 UTC; 6s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 6365 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 6368 (apache2)
     Tasks: 55 (limit: 4999)
    Memory: 0.3M (peak: 8.6M)
       CPU: 35ms
   CGroup: /system.slice/apache2.service
           └─6368 /usr/sbin/apache2 -k start
             └─6370 /usr/sbin/apache2 -k start
               └─6371 /usr/sbin/apache2 -k start

Aug 24 03:46:55 IT-12 systemd[1]: Starting apache2.service - The Apache HTTP Server.
Aug 24 03:46:55 IT-12 systemd[1]: Started apache2.service - The Apache HTTP Server.
mvlu@IT-12: /etc/ssl/mycert$ openssl s_client -connect localhost:443
CONNECTED(00000003)
Can't use SSL_get_servername
depth=0 C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = rathodunnatics242541@gmail.com
verify error:num=18:self-signed certificate
verify return:1
depth=0 C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = rathodunnatics242541@gmail.com
verify return:1
---
Certificate chain
```

Step 8: Configure Apache for HTTPS

```
mvl@IT-12:/etc/ssl/mycert %  
Extended master secret: no  
Max Early Data: 0  
  
-----  
read R BLOCK  
^C  
mvl@IT-12:/etc/ssl/mycert$ openssl x509 -in /etc/ssl/mycert/server.crt -text -noout  
Certificate:  
Data:  
Version: 3 (0x2)  
Serial Number:  
56:e9:a1:9c:17:3b:9c:5b:16:40:40:4f:47:ed:4f:87:9a:b8:27:90  
Signature Algorithm: sha256WithRSAEncryption  
Issuer: C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, emailAddress = rathodunnatics242541@gmail.com  
Validity  
Not Before: Aug 24 03:48:51 2026 GMT  
Not After : Aug 24 03:48:51 2027 GMT  
Subject: C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = rathodunnatics242541@gmail.com  
Subject Public Key Info:  
Public Key Algorithm: rsaEncryption  
Public-Key: (2048 bit)  
Modulus:  
00:b4:e4:c7:38:a6:ae:89:6e:d2:9c:e1:96:57:e2:  
d8:3a:e4:1f:bf:da:3d:9a:25:7f:6a:f5:33:10:f4:  
f0:7c:f5:15:dd:2d:b0:71:3f:28:01:92:e7:52:8a:  
b6:6d:5b:98:66:40:1a:5e:9f:29:40:c5:37:d1:e2:  
cc:d0:70:24:fa:65:3c:ee:e6:8c:62:0d:a1:be:4e:  
01:a4:f9:18:9f:9e:11:a7:70:22:09:d4:00:0b:55:  
5f:a5:81:09:9b:6a:a4:b4:41:f5:d5:ae:72:31:48:  
a9:d6:d3:a0:ba:54:97:c2:9e:16:8c:d6:df:f9:5f:  
f6:84:66:4e:ca:f9:a8:2a:9d:a0:23:ee:76:c8:f7:  
8b:67:82:db:9c:a9:ed:0e:2d:4b:72:34:e0:5c:8e:  
70:e8:db:09:20:6d:34:83:50:e0:d5:cc:92:23:8c:  
06:8d:ec:ec:d9:42:a7:ef:59:16:f6:64:82:6f:81:  
1e:69:40:5a:41:30:a4:11:3f:77:42:9c:6d:29:d4:  
f6:6e:a4:06:01:0b:49:31:20:a3:30:4a:8a:c7:32:  
fe:a3:34:b1:ee:76:de:fc:18:9f:85:d9:9f:9c:7d:  
07:fb:5c:ab:a8:01:71:40:c7:fe:61:32:97:c7:15:  
bb:01:5e:2c:d7:26:a7:a4:74:4e:e6:3d:9a:e9:ed:  
f3:67  
Exponent: 65537 (0x10001)  
X509v3 extensions:  
X509v3 Subject Key Identifier:
```

Step 9: Test Apache Configuration

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```
mvsu@IT-12:/etc/ssl/mycert$ mvlu@IT-12:/etc/ssl/mycert$ sudo nano /etc/apache2/mods-available/ssl.conf
mvlu@IT-12:/etc/ssl/mycert$ sudo systemctl restart apache2
mvlu@IT-12:/etc/ssl/mycert$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-08-24 04:00:15 UTC; 5s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 6964 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
 Main PID: 6967 (apache2)
    Tasks: 55 (limit: 4499)
   Memory: 8.1M (peak: 8.2M)
      CPU: 34ms
   CGroup: /system.slice/apache2.service
           └─6967 /usr/sbin/apache2 -k start
             └─6969 /usr/sbin/apache2 -k start
               └─6970 /usr/sbin/apache2 -k start

Aug 24 04:00:15 IT-12 systemd[1]: Starting apache2.service - The Apache HTTP Server.
Aug 24 04:00:15 IT-12 systemd[1]: Started apache2.service - The Apache HTTP Server.
mvsu@IT-12:/etc/ssl/mycert$ openssl s_client -connect localhost:443
CONNECTED(00000003)
Can't use SSL_get_servername
depth=0 C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = rathodunnatics242541@gmail.com
verify error:num=18:self-signed certificate
verify return:1
depth=0 C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = rathodunnatics242541@gmail.com
verify return:1
---
Certificate chain
 0 s:C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = rathodunnatics242541@gmail.com
  i:C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = rathodunnatics242541@gmail.com
 a:PKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256
 v:NotBefore: Aug 24 03:40:51 2026 GMT; NotAfter: Aug 24 03:40:51 2027 GMT

Server certificate
-----BEGIN CERTIFICATE-----
MIIEKzCCAoQgAIBAgIUUmhn8C7jFsWQEBPRr1Ph5q4J3AwDQYJKoZIhvcNAQEL
BQAwAgQxChA3BgNVBAYTAkwwRQwEgYDVQIDAtNVMhhcmFzaHRYdTEPMAGAA1UE
BwwGTXYmFpNRMeEQYDVQQDApNeS8Db2xsZXRwYWZlYDVOQ0QDAUJDUyBEZXhh
cnRnTSW5MHRlWEAYDQOQDAsb2hhbGhvc3QxLTAuRzBkghkiG9w8BCUHMhJdGhd
ZHVubmF0aWZmMjQyNTQxQ0GdYVJlLmNmN2BwFwYjAUMjQ1MzQwMjUwNzA0
MjQwMzQwNTFAhGJGkMqSwCQDVQ0QGEwJjJTEUMBTGA1UECAwLTwFoYXJ0c2h0cmEx
-----
```

Step 10: Restart Apache

Step 11: Allow HTTPS Through Firewall

```
mwlw@IT-12: /etc/ssl/mycert$ x
2f:65:bf:e8:32:63:53:7d:30:6a:63:d3:19:e6:31:6a:cd:2b:
9b:6a:57:ac:04:21:75:eb:4a:c5:8e:ae:7a:f4:b0:5a:a7:a2:
ed:6f:6c:ae:25:ef:a6:63:0f:aa:7a:20:81:74:c8:a9:97:e4:
21:37:6b:a7:61:b9:e8:5f:1d:1f:14:66:56:aa:7b:47:7f:cf:
78:65:83:e1
mwlw@IT-12: /etc/ssl/mycert$ openssl x509 -enddate -noout -in /etc/ssl/mycert/server.crt
notAfter=Aug 24 03:40:51 2027 GMT
mwlw@IT-12: /etc/ssl/mycert$ sudo nano /etc/apache2/mods-available/ssl.conf
mwlw@IT-12: /etc/ssl/mycert$ sudo systemctl restart apache2
mwlw@IT-12: /etc/ssl/mycert$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enable)
   Active: active (running) since Mon 2026-08-24 04:00:15 UTC; 5s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 6964 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Main PID: 6967 (apache2)
       Tasks: 55 (limit: 4499)
      Memory: 8.1M (peak: 8.2M)
         CPU: 34ms
    CGroup: /system.slice/apache2.service
            └─6967 /usr/sbin/apache2 -k start
              └─6969 /usr/sbin/apache2 -k start
                └─6970 /usr/sbin/apache2 -k start

Aug 24 04:00:15 IT-12 systemd[1]: Starting apache2.service - The Apache HTTP Server.
Aug 24 04:00:15 IT-12 systemd[1]: Started apache2.service - The Apache HTTP Server.
mwlw@IT-12: /etc/ssl/mycert$ openssl s_client -connect localhost:443
CONNECTED(000000003)
Can't use SSL_get_servername
depth=0 C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = rathodunnatics242541@gmail.com
verify error:num=18:self-signed certificate
verify return:1
depth=0 C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = rathodunnatics242541@gmail.com
verify return:1

Certificate chain
 0 s:C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = rathodunnatics242541@gmail.com
 1 i:C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = rathodunnatics242541@gmail.com
 a:PKKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256
 v:NotBefore: Aug 24 03:40:51 2026 GMT, NotAfter: Aug 24 03:40:51 2027 GMT

-----
Server certificate
```

Step 13: Verify the TLS Connection

Step 14: Check the Certificate

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```
mdu@mtl-07:/etc/balmycert$ cd ..  
Main PID: 6871 (apache2)  
Tasks: 55 (limit: 4499)  
Memory: 8.1M (peak: 8.4M)  
CPU: 37ms  
CGroup: /system.slice/apache2.service  
├─6871 /usr/sbin/apache2 -k start  
├─6873 /usr/sbin/apache2 -k start  
└─6874 /usr/sbin/apache2 -k start
```

Aug 24 03:49:47 IT-12 systemd[1]: Starting apache2.service - The Apache HTTP Server.
Aug 24 03:49:47 IT-12 systemd[1]: Started apache2.service - The Apache HTTP Server.

```
mvlw@IT-12:/etc/ssl/mycert$ sudo ufw status  
sudo: ufw: command not found  
mvlw@IT-12:/etc/ssl/mycert$ sudo ufw allow 443/tcp  
sudo: ufw: command not found  
mvlw@IT-12:/etc/ssl/mycert$ openssl s_client -connect localhost:443  
CONNECTED(00000003)  
Can't use SSL_get_servername  
depth=0 C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = rathodunnatics242541@gmail.com  
verify error:num=18:self-signed certificate  
verify return:1  
depth=0 C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = rathodunnatics242541@gmail.com  
verify return:1
```

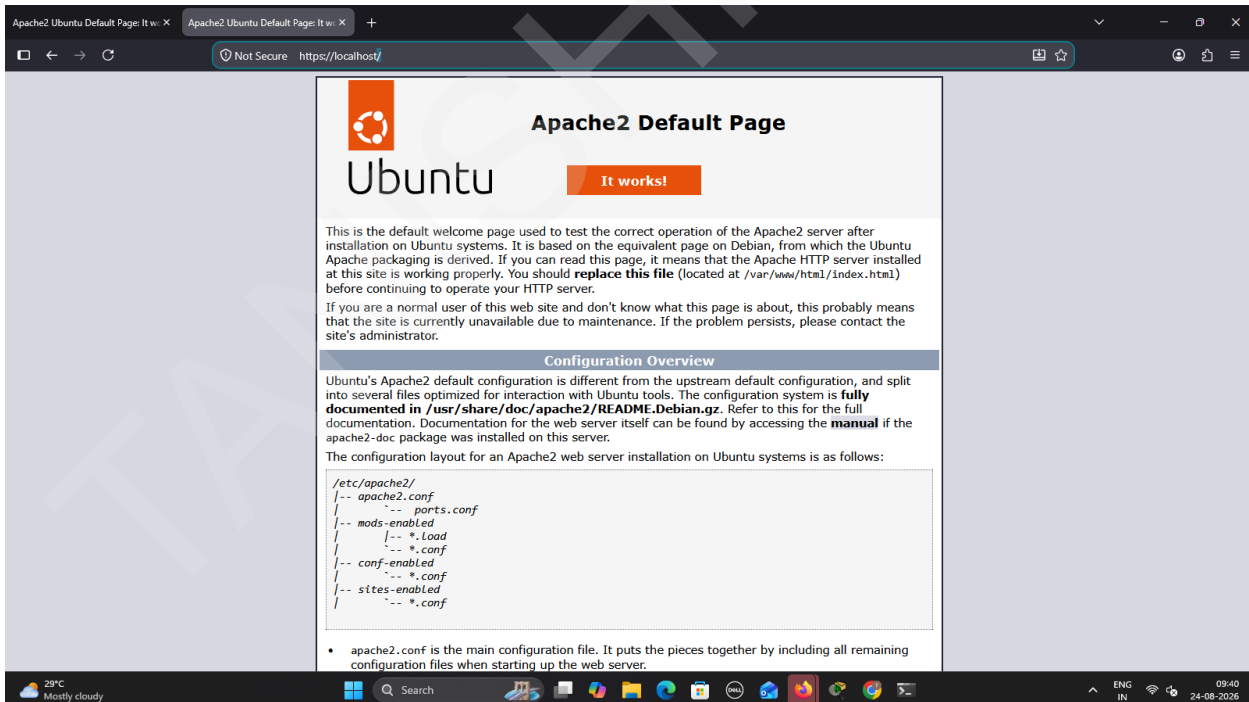
Certificate chain

```
o=C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = rathodunnatics242541@gmail.com  
i:C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = rathodunnatics242541@gmail.com  
a:PKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256  
v:NotBefore: Aug 24 03:40:51 2026 GMT; NotAfter: Aug 24 03:40:51 2027 GMT
```

Server certificate

```
-----BEGIN CERTIFICATE-----  
MIIEKzCCAxOgAwIBAgIUUVumhnBc7JfSwQEBrPR+1PhSq4J5AmDQYJKoZIhvcNAQEL  
BgQAAQCxZCAABglVBAAYTAHLOHRQWegYDVQQIDATNYWhhcFZhYyRTEPMAAGAUEU  
bmVudGVyaXNpbGluZW5kaSBBMBEGBjAUBgcqhkIIGDAIUDlgBYeZVBh  
cnRTMSMRLEIAVDQDDALab2hhbGlzc3xlTAEBlghqhkiG9w0BBQCEWHnjdhdChv  
ZHViubGFoaWlnZjhjoYTQocdvtYdlLsmLvbtAEFAbyNJAJMJQMztFWtFaWobyNZAU  
IJMQMZQuNTfaMTGWQswCOYDUVOQGEGJJTJEUMBIGAIUECAMLTlfFoYXHjc2bcEmEx  
DzANBGNVBAcMBKA1blbJhaTETMBEGA1UECgwTKkg029slsbGVnZVNzTEMBOGA1UECWwN  
IQ1MGRGVuYXJBbmVudDESMBAGAIUAewJbg6YJmxbob3NMGSOMhwUKJoKOZIhvcNAQkB  
Ph5YYRob2RIbmShdGIJCzMjIMjUHUBNBWFpbCc5jb2BwggeEAEMGCSCQSgsTB3DOEB  
AQAAAAIBDWawggEKAAQIBAQCOCBMcl4pq6bjtkCUZZxtGtg6SB+/2j2aZX9g9TMQP9PB8
```

Step 15: Secure Certificate Management



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```
mvl@IT-12: /etc/ssl/mycert$ openssl s_client -connect 127.0.0.1:443
Peer signing digest: SHA256
Peer signature type: RSA-PSS
Server Temp Key: X25519, 253 bits
---
SSL handshake has read 1627 bytes and written 373 bytes
Verification error: self-signed certificate
---
New, TLSv1.3, Cipher is TLS_AES_256_GCM_SHA384
Server public key is 2048 bit
Secure Renegotiation IS NOT supported
Compression: NONE
Expansion: NONE
No ALPN negotiated
Early data was not sent
Verify return code: 18 (self-signed certificate)
---
Post-Handshake New Session Ticket arrived:
SSL-Session:
    Protocol  : TLSv1.3
    Cipher    : TLS_AES_256_GCM_SHA384
    Session-ID: E10B5D72B0D832F80FE67B8F74D852D3D3F974FBCFA3C33DAB70B14789C1E7
    Session-ID-ctx:
    Resumption PSK: EB02A4ABE4708AAE78D05881411E0910549EF37649E2F8AA0E689FA148903A47986CFEED743F23413258F377E653B1C7
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    TLS session ticket lifetime hint: 300 (seconds)
    TLS session ticket:
    0000 - 07 a4 1a f6 d3 1a 59 75-24 a6 d4 02 d5 f8 49 90 .....Yu$.....I.
    0010 - 58 c5 af 11 ed 6d 1f 0f-15 c3 20 56 90 79 b4 3e X....m....V.y.>

    Start Time: 1787543743
    Timeout    : 7200 (sec)
    Verify return code: 18 (self-signed certificate)
    Extended master secret: no
    Max Early Data: 0
---
read R BLOCK
---
Post-Handshake New Session Ticket arrived:
SSL-Session:
```

Step 16: Configure Strong TLS Security

```
mvl@IT-12: /etc/ssl/mycert$ openssl x509 -enddate -noout -in /etc/ssl/mycert/server.crt
f3:67
Exponent: 65537 (0x10001)
X509v3 extensions:
    X509v3 Subject Key Identifier:
        7D:05:CF:0D:E8:61:51:89:A4:71:7D:E9:CC:6F:A2:0E:10:62:AC:CE
    X509v3 Authority Key Identifier:
        7D:05:CF:0D:E8:61:51:89:A4:71:7D:E9:CC:6F:A2:0E:10:62:AC:CE
    X509v3 Basic Constraints: critical
        CA:TRUE
Signature Algorithm: sha256WithRSAEncryption
Signature Value:
    16:d6:bd:55:29:27:a8:a7:6f:4d:74:5e:a7:b2:41:18:9d:
    d6:3c:36:6c:11:25:93:42:fc:d8:47:3e:e2:a1:0c:59:3d:a3:
    d6:d1:34:a2:06:e0:ce:3a:a2:be:f5:a0:15:30:d7:0f:e9:cd:
    d1:e6:11:e9:c5:0e:5a:0e:b1:09:08:d7:83:aa:3e:a7:2f:cb:
    4b:0b:38:55:39:b7:ac:d7:cb:dc:fd:7f:cc:6e:26:31:a7:1b:
    91:22:07:fb:c3:f5:c1:44:b6:30:c9:e4:3f:7b:76:2a:bf:0e:
    93:6f:9a:c5:0e:9e:df:c6:fd:c5:4e:d6:28:c7:64:82:62:a0:
    86:75:db:1e:84:51:ac:e7:65:14:99:f5:7e:ab:33:d8:9b:a6:
    91:e0:a1:f2:c5:e1:1a:2d:05:8d:96:42:64:e1:f5:56:9f:5e:
    a1:f9:b0:be:fa:4f:e4:b9:3d:83:8f:07:88:88:d0:ce:30:62:
    2f:65:bf:e8:32:63:53:7d:30:6a:63:d3:19:e6:31:6a:cd:2b:
    9b:6a:57:ac:04:21:75:eb:4a:c5:8e:ae:7a:f4:b0:5a:a7:a2:
    ed:6f:6c:ae:25:ef:a6:63:0f:aa:7a:20:81:74:c8:a9:97:e4:
    21:37:6b:a7:61:b9:e8:5f:1d:1f:14:66:56:aa:7b:47:7f:c4:
    78:65:83:e1
mvl@IT-12: /etc/ssl/mycert$ openssl x509 -enddate -noout -in /etc/ssl/mycert/server.crt
notAfter=Aug 24 03:40:51 2027 GMT
mvl@IT-12: /etc/ssl/mycert$ sudo nano /etc/apache2/mods-available/ssl.conf
mvl@IT-12: /etc/ssl/mycert$ sudo systemctl restart apache2
mvl@IT-12: /etc/ssl/mycert$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-08-24 04:00:15 UTC; 5s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 6964 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 6967 (apache2)
     Tasks: 55 (Limit: 4499)
   Memory: 8.1M (peak: 8.2M)
      CPU: 34ms
   CGroup: /system.slice/apache2.service
           └─6967 /usr/sbin/apache2 -k start
```

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Expected Result

After completing the practical:

- Apache web server is configured.
- SSL/TLS is enabled.
- A digital certificate and private key are generated.
- HTTPS communication is established on port 443.

- The TLS session can be verified using OpenSSL.
- Certificate information and expiry can be managed.
- Communication between browser and server is encrypted.